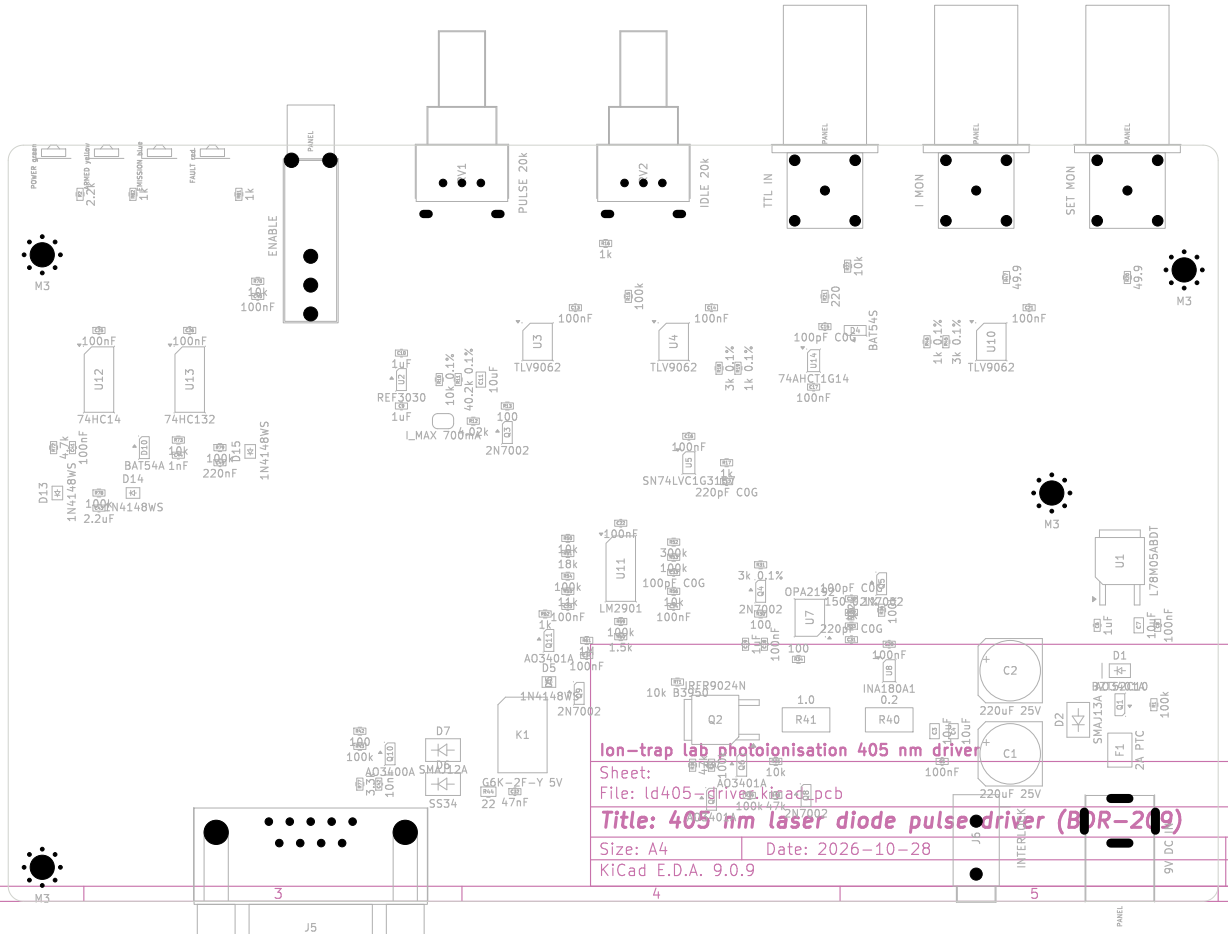

POWER green
ARMED yellow
EMISSION blue
FAULT red
ENABLE
PULSE 20k
IDLE 20k
TTL IN
I MON
SET MON
PANEL
M3
U12
74HC14
U13
74HC132
U3
TLV9062
U4
TLV9062
U10
TLV9062
U11
LM2901
U7
OPA2188
U1
L78M05ABDT
REF3030
L_MAX 700mA
2N7002
SN74LVC1G3157
74AHCT1G14
BAT54S
BA154A
1N4148WS
AO3401A
IRFR9024N
INA180A1
K1
G6K-2F-Y 5V
SS34
SMAJ12A
SMAJ13A
BZX84C10
Q2
R41
R40
C2
220uF 25V
C1
220uF 25V
D1
D2
F1
2A PTC
J5
LD OUT
INTERLOCK
9V DC
Ion-trap lab photoionisation 405 nm driver
Sheet:
File: ld405_driver_knacq.kicad_pcb
Title: 405 nm laser diode pulse driver (BDR-219)
Size: A4
Date: 2026-10-28
KiCad E.D.A. 9.0.9
Rev: 1.0
Id: 1/1